Jacobian-Velocity Bounds for Deployment Risk Under Covariate Drift

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***Abstract*—We study long-horizon deployment of a frozen predictor under dynamic covariate shift. A time-domain Poincare inequality reduces temporal risk volatility to derivative energy, and a Jacobian-velocity theorem identifies directional tangent energy along the deployment path as the governing quantity under explicit along-path regularity and domination assumptions. Under low-rank drift, that quantity reduces to directional Jacobian energy in the drift subspace, motivating drift-aligned tangent regularization (DTR) and a matched monitoring proxy. Rather than smoothing the network isotropically, DTR penalizes sensitivity only along estimated drift directions. We validate the theorem-to-method pipeline in three experiments, namely a synthetic benchmark for the time-domain inequality, a controlled synthetic comparison against isotropic Jacobian regularization, and a field-deployment study on the UCI Air Quality sensor dataset. DTR consistently reduces risk volatility and directional gain relative to standard training and isotropic smoothing. Moderate drift-subspace misspecification is tolerable while orthogonal misspecification largely removes the benefit.**

***Index Terms*—covariate drift, deployment risk, Jacobian regularization, temporal robustness, distribution shift**

I. Introduction

Distribution shift is often discussed statically by training on one distribution, testing on another, and measuring the gap [1], [2]. Long-horizon deployment is different. A model may be frozen while the environment moves continuously, so the relevant object is not a single out-of-domain error but a risk trajectory. Temporal benchmarks and deployment studies make the same point from different angles. Even when labels and architecture are fixed, performance can move materially over time [3]–[6]. The practical question is therefore not only whether shift occurs, but how violently performance can move once deployment begins.

This paper develops a geometric answer. If the feature stream follows a path X_t and the predictor has local tangent map $J_f(X_t)$, then instability is governed not by shift magnitude alone but by the directional interaction between the data velocity $\dot{X}_t$ and the model tangent geometry. The dangerous quantity is $J_f(X_t)\dot{X}_t$. The intuition is simple. The same deployment path can be almost harmless for one model and damaging for another, because only motion through steep tangent directions accumulates into large score variation. If the stream moves through a flat direction, drift can be present without being operationally dangerous. If it moves through a steep direction, even gentle but persistent motion can compound into long-horizon degradation. The paper's central bound formalizes this. Under explicit along-path regularity and directional domination assumptions,

$$\mathrm{Var}_U(r(U)) \leq \frac{\beta^2 T}{\pi^2} \int_0^T \mathbb{E}\|J_f(X_t)\dot{X}_t\|^2 dt.$$

First, a time-domain Poincare inequality converts temporal volatility into derivative energy. Second, a Jacobian-velocity theorem bounds that energy under explicit along-path assumptions on the deployment performance field. Third, a low-rank drift specialization naturally motivates a drift-aligned regularizer and a matched monitoring proxy. Fourth, we test the resulting story with a synthetic time-domain sanity check, a compact isotropic-versus-directional comparison, and a real field-deployment study on Air Quality. The empirical sequence mirrors the logical one. It first verifies the temporal inequality itself, then compares competing regularizers in the idealized low-rank regime, and only then moves to a real deployment path where the drift subspace must be estimated. Concretely, the paper identifies directional tangent energy as the quantity governing frozen-model deployment volatility, uses the resulting low-rank reduction to motivate drift-aligned tangent regularization (DTR) and a matched hazard proxy, and gives empirical evidence that the same directional geometry matters in both controlled synthetic experiments and a real field-deployment study. DTR reduces risk volatility and directional gain relative to standard training and isotropic smoothing.

The paper is narrow by design. It does not address general distribution shift or test-time adaptation, but asks specifically what local geometry governs frozen-model degradation and whether that geometry remains visible on real drifting data. Code, scripts, and supplementary materials are available in a public repository [7]. A companion project page with figures and summaries is available at [8].

II. Related Work

The paper sits closest to temporal distribution shift and dynamic deployment evaluation. Classical dataset-shift taxonomies separate covariate, label, and concept shift [1], [2], while concept-drift work emphasizes sequential adaptation under evolving streams [3]. More recent benchmark-driven work has made temporal structure explicit, both in naturally shifted datasets and in stylized train-deploy protocols [4]–[6], [9]. Our focus is narrower. Rather than estimating a worst-case

shift gap, we study the volatility of one frozen model along a deployment path.

The paper is also adjacent to test-time adaptation [10]–[12]. Those methods update the model online to track changing data. Here the predictor remains frozen. That restriction is deliberate. It isolates the geometric mechanism of degradation itself and yields a quantity that can be analyzed, regularized, and monitored without mixing in the dynamics of online parameter updates.

Finally, the work draws on two technical lines. The regularization side is related to Jacobian-based control of sensitivity and robustness [13]–[15], but the object here is directional rather than isotropic. Only tangent energy in likely drift directions enters the bound. The monitoring side is related to shift detection, uncertainty under shift, and sequential change detection [16]–[22]. Our hazard score is not presented as a sequentially optimal detector. It is a model-aware proxy matched to the same drift geometry used in training.

III. Setup and Assumptions

Let $t \in [0, T]$ index deployment time. Let $X_t \in \mathbb{R}^d$ denote the covariate process, and let $f_\theta : \mathbb{R}^d \to \mathbb{R}$ be a frozen scalar score function. We study a scalar deployment performance field $g_\theta : \mathbb{R}^d \to \mathbb{R}$ and the associated trajectory

$$r(t) := \mathbb{E}[g_\theta(X_t)]. \tag{1}$$

For classification, a natural example is

$$g_\theta(x) = \mathbb{E}_{Y \sim P_0(\cdot|x)}[\ell(f_\theta(x), Y)],$$

but the theorem below does not rely on this representation alone. Instead, we state the regularity needed to make the risk-trajectory argument defensible. The point of this setup is to separate two roles that are often conflated in shift analyses. The model is frozen, while the covariate stream is allowed to move.

To measure temporal instability, draw $U \sim \mathrm{Unif}[0, T]$ and define

$$\mathrm{Var}_U(r(U)) = \mathbb{E}[(r(U) - \mathbb{E}r(U))^2]. \tag{2}$$

The quantity in (2) is the notion of deployment volatility studied throughout the paper.

Assumption A1 (trajectory regularity). The sample paths $t \mapsto X_t$ are almost surely absolutely continuous and

$$\mathbb{E} \int_0^T \|\dot{X}_t\|^2 dt < \infty.$$

Assumption A2 (chain rule under the expectation). The field g_θ is weakly differentiable, $t \mapsto g_\theta(X_t)$ is almost surely absolutely continuous, and

$$g_\theta(X_t) - g_\theta(X_s) = \int_s^t \nabla g_\theta(X_u)^\top \dot{X}_u \, du,$$

for all $0 \le s < t \le T$ on almost every sample path, with

$$\mathbb{E} \int_0^T |\nabla g_\theta(X_t)^\top \dot{X}_t| dt < \infty.$$

Assumption A3 (score Jacobian well posed and directional domination along deployment). There exists a measurable matrix field J_f such that $J_f(X_t)$ is well defined almost surely for almost every $t \in [0, T]$,

$$\mathbb{E} \int_0^T \|J_f(X_t)\dot{X}_t\|^2 dt < \infty,$$

and there exists $\beta > 0$ such that

$$|\nabla g_\theta(X_t)^\top \dot{X}_t| \le \beta \|J_f(X_t)\dot{X}_t\|$$

for almost every $t \in [0, T]$ on almost every sample path.

Assumption A3 is the paper's main compatibility condition. It should not be read as a consequence of dynamic covariate shift alone. Rather, it asserts that along the realized deployment path, the local change in the chosen performance field is dominated by the local change in the score, and that the score Jacobian is well posed there.

Remark 1 (What A3 means, when it holds, and when it fails)**.** *A3 is deliberately along-path rather than pointwise in every ambient direction. The proof only needs control of $\nabla g_\theta(X_t)^\top \dot{X}_t$ by $J_f(X_t)\dot{X}_t$ along the realized velocity.*

A3 holds in the natural composition case. If $g_\theta(x) = h(f_\theta(x))$, then

$$\nabla g_\theta(X_t)^\top \dot{X}_t = h'(f_\theta(X_t))\, J_f(X_t)\dot{X}_t,$$

so A3 follows with $\beta = \sup |h'|$ whenever $J_f(X_t)$ is well defined almost surely for almost every t. For Bernoulli cross-entropy against a fixed soft target, $|h'| \le 1$, giving $\beta = 1$. The theorem is therefore not ReLU-specific. Smooth predictors fit directly, and piecewise-linear predictors such as ReLU networks fit as soon as the deployment path avoids their nondifferentiability set almost surely for almost every deployment time.

A3 does not follow from pure covariate shift alone. The general risk $g_\theta(x) = \mathbb{E}_{Y \sim P_0(\cdot|x)}[\ell(f_\theta(x), Y)]$ carries an extra $\nabla \eta(x)$ term from the label conditional, which need not be dominated by the score-Jacobian term even when η is time-invariant. A3 also fails under simultaneous concept shift, and at score saturation wherever $\nabla f_\theta(x) = 0$ but $\nabla g_\theta(x) \neq 0$.

IV. Main Theorem Package

The first step is purely temporal.

Lemma 1 (Time-domain derivative-energy control)**.** *If r is absolutely continuous on $[0, T]$, then*

$$\mathrm{Var}_U(r(U)) \le \frac{T}{\pi^2} \int_0^T (r'(t))^2 dt. \tag{3}$$

Proof. Let $\bar{r} = T^{-1} \int_0^T r(t)\, dt$. The Wirtinger inequality yields

$$\int_0^T (r(t) - \bar{r})^2 dt \le \frac{T^2}{\pi^2} \int_0^T (r'(t))^2 dt. \tag{4}$$

Divide by T. □

Lemma 1 says volatility is impossible without derivative energy; see (3). The next result identifies the geometric driver of that energy.

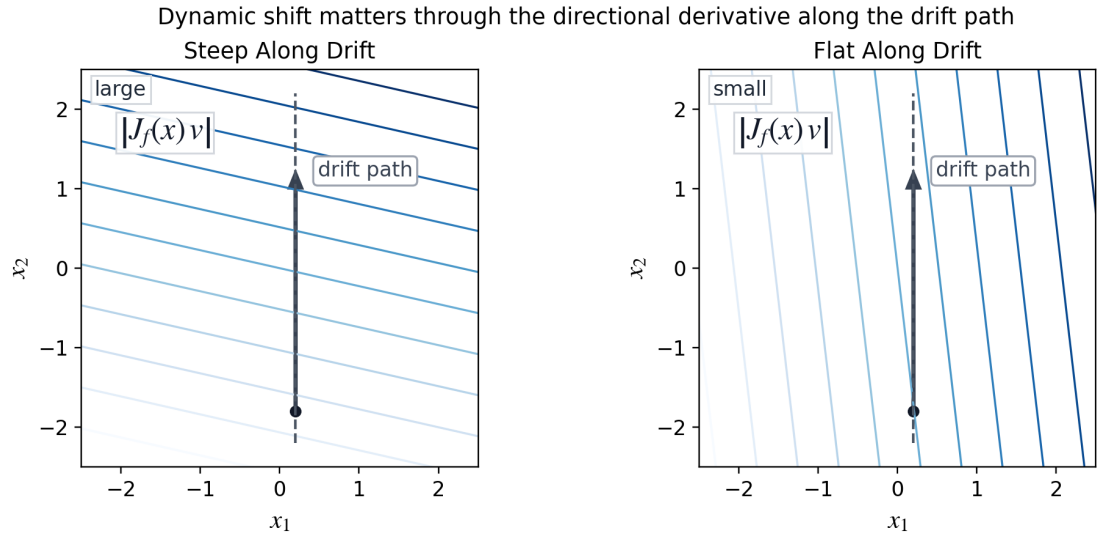

Fig. 1. Geometric intuition for drift-aligned instability. The deployment path is the same in both panels, but the local tangent geometry is different. When the drift direction aligns with a steep tangent direction, the score changes rapidly, whereas when the model is flat along that same direction, the score remains stable. Figure 1 visualizes the directional quantity that appears in Theorem 1 and (5).

Theorem 1 (Jacobian-velocity control of risk volatility)**.** *Assume A1–A3. Then r is absolutely continuous and*

$$\mathrm{Var}_U(r(U)) \le \frac{\beta^2 T}{\pi^2} \int_0^T \mathbb{E}\|J_f(X_t)\dot{X}_t\|^2 dt. \tag{5}$$

Proof. Fix $0 \le s < t \le T$. By Assumption A2 and Fubini,

$$\begin{aligned} r(t) - r(s) &= \mathbb{E}\left[\int_s^t \nabla g_\theta(X_u)^\top \dot{X}_u \, du\right] \\ &= \int_s^t \mathbb{E}[\nabla g_\theta(X_u)^\top \dot{X}_u] \, du. \end{aligned}$$

Hence r is absolutely continuous and, for almost every t,

$$r'(t) = \mathbb{E}[\nabla g_\theta(X_t)^\top \dot{X}_t]. \tag{6}$$

Jensen's inequality and Assumption A3 then give

$$\begin{aligned} (r'(t))^2 &\le \mathbb{E}[(\nabla g_\theta(X_t)^\top \dot{X}_t)^2] \\ &\le \beta^2 \mathbb{E}\|J_f(X_t)\dot{X}_t\|^2. \end{aligned} \tag{7}$$

Apply Lemma 1 and (3). □

Theorem 1 isolates a single instability mechanism for the trajectory in (1). Deployment volatility is driven by accumulated tangent amplification along the realized motion of the data stream, as captured by (5). Figure 1 is a geometric rendering of that statement. In that sense, the theorem turns Jacobian-based sensitivity control [13]–[15] into a statement about temporal deployment risk rather than static local robustness. The bound also has a useful operational reading. Volatility requires two ingredients at once, data motion and tangent gain. If either one is small, long-horizon fluctuation remains small.

The low-rank regime gives the algorithmic specialization.

Corollary 1 (Drift-subspace bound)**.** *Assume the hypotheses of Theorem 1. If*

$$\dot{X}_t = V a_t + \rho_t, \qquad V \in \mathbb{R}^{d\times k},$$

with $V^\top V = I_k$ and $V^\top \rho_t = 0$, then

$$\mathrm{Var}_U(r(U)) \le \frac{2\beta^2 T}{\pi^2}(B_V + B_\rho), \tag{8}$$

where

$$\begin{aligned} B_V &:= \int_0^T \mathbb{E}[\|J_f(X_t)V\|_F^2 \|a_t\|^2] dt, \\ B_\rho &:= \int_0^T \mathbb{E}\|J_f(X_t)\rho_t\|^2 dt. \end{aligned}$$

Proof. By Theorem 1 and (5),

$$\begin{aligned} \|J_f(X_t)\dot{X}_t\|^2 &= \|J_f(X_t)Va_t + J_f(X_t)\rho_t\|^2 \\ &\le 2\|J_f(X_t)Va_t\|^2 + 2\|J_f(X_t)\rho_t\|^2. \end{aligned}$$

Since V has orthonormal columns,

$$\|J_f(X_t)Va_t\|^2 \le \|J_f(X_t)V\|_F^2 \|a_t\|^2.$$

Integrate over t. □

Corollary 1 is the paper's main reduction. When B_ρ is small, the leading stability term in (8) is governed by the directional Jacobian energy in the drift subspace rather than by isotropic smoothness. This is precisely the regime in which low-dimensional temporal drift and low-rank change models are empirically plausible [5], [20].

V. Method

The theorem package suggests a simple design principle. If instability is created by motion through steep directions, then regularization should flatten the model along directions deployment is expected to traverse, not everywhere at once.

A. Drift-aligned tangent regularization

Let V estimate the dominant drift subspace from unlabeled deployment covariates. Two minimal estimators are sufficient for the present paper. One is a mean-difference direction

$$\Delta\mu_t = \mu_t - \mu_{t-\Delta}, \qquad v_t = \frac{\Delta\mu_t}{\|\Delta\mu_t\|},$$

or the top-k principal directions of the difference cloud $\{X_t - X_{t-\Delta}\}$ over a rolling window.

With a fixed subspace estimate V, we train using

$$\mathcal{L}_{\mathrm{DTR}}(\theta) = \mathbb{E}_{(X,Y)}\ell(f_\theta(X), Y) \;+\; \lambda\mathbb{E}_X\|J_f(X)V\|_F^2. \quad (9)$$

The objective in (9) is not global Jacobian smoothing. It is a directional sensitivity constraint. The model is flattened only along directions expected to dominate future drift. Corollary 1 explains why this is the quantity worth shrinking, and it distinguishes DTR from isotropic Jacobian penalties that aim at broader smoothness or margin control [13]–[15]. In other words, DTR concentrates regularization budget on directions the future trajectory is expected to traverse, rather than spreading it uniformly over directions the deployment path may never visit.

B. Monitoring score

The same geometry yields a matched deployment diagnostic, now at the level of a theorem-motivated proxy rather than the literal corollary integrand. Let

$$\begin{aligned} s_t &:= \|\Delta\mu_t\|/\Delta, \quad g_t := \mathbb{E}\|J_f(X_t)V_t\|_F^2, \\ h_t &:= s_t^2 g_t. \end{aligned} \quad (10)$$

Proposition 1 (Rank-1 bookkeeping for the hazard score)**.** *Consider the rank-1 monitoring proxy used below, so that $V_t = v_t \in \mathbb{R}^d$ is a unit vector. Suppose the block mean difference satisfies*

$$\frac{\Delta\mu_t}{\Delta} = v\bar{a}_t + \bar{\rho}_t, \qquad \|v\| = 1, \qquad v^\top\bar{\rho}_t = 0,$$

where $\bar{a}_t$ is the block-averaged aligned drift coefficient and $\bar{\rho}_t$ is the block-averaged residual drift. Write

$$v_t = \cos\theta_t\, v + \sin\theta_t\, u_t, \qquad u_t^\top v = 0, \qquad \|u_t\| = 1,$$

and define

$$\begin{aligned} G_{\|,t} &:= \mathbb{E}\|J_f(X_t)v\|^2, \\ G_{\perp,t} &:= \mathbb{E}\|J_f(X_t)u_t\|^2, \\ C_t &:= \mathbb{E}\langle J_f(X_t)v, J_f(X_t)u_t\rangle. \end{aligned}$$

Then

$$s_t^2 = |\bar{a}_t|^2 + \|\bar{\rho}_t\|^2$$

and

$$g_t = \cos^2\theta_t\, G_{\|,t} \;+\; \sin^2\theta_t\, G_{\perp,t} \;+\; 2\sin\theta_t\cos\theta_t\, C_t.$$

Consequently,

$$\begin{aligned} h_t = {} & (|\bar{a}_t|^2 + \|\bar{\rho}_t\|^2) \\ & \times \Big(\cos^2\theta_t\, G_{\|,t} \;+\; \sin^2\theta_t\, G_{\perp,t} \;+\; 2\sin\theta_t\cos\theta_t\, C_t\Big). \end{aligned}$$

Proof. The identities are immediate from orthogonal decompositions. The first uses $(\Delta\mu_t)/\Delta = v\bar{a}_t + \bar{\rho}_t$ with $v^\top\bar{\rho}_t = 0$. The second expands $J_f(X_t)v_t = \cos\theta_t\, J_f(X_t)v + \sin\theta_t\, J_f(X_t)u_t$ and takes expectations. □

Proposition 1 makes the proxy gap explicit in the rank-1 monitoring setting. The finite-window drift statistic contributes the residual term $\|\bar{\rho}_t\|^2$, while angular error mixes aligned, orthogonal, and overlap energies through θ_t, $G_{\perp,t}$, and C_t. In that precise bookkeeping sense, h_t matches the leading low-rank quantity from Corollary 1 up to block averaging, residual drift, and subspace-estimation error rather than reproducing the corollary integrand exactly. It becomes large only when the stream is moving quickly *and* the predictor is steep along that motion. Relative to generic shift alarms [16], [17] and sequential detection procedures [18], [19], [22], the point of h_t is alignment rather than optimality. It monitors the same directional mechanism singled out by the theory without claiming to be a sequentially optimal detector. This symmetry between training and monitoring is deliberate. The same geometry used to suppress future instability is reused to measure when future instability is likely.

VI. Experiments

A. Synthetic time-domain sanity check

The synthetic study is fully specified and uses the smallest setup that exposes the paper's mechanism. There is one stable signal coordinate, one drifting nuisance coordinate, and a frozen classifier whose tangent geometry can either amplify or suppress that drift. This kind of temporally evolving evaluation is consistent with recent temporal-shift benchmarks, but here the drift direction is deliberately known so the theorem-to-method link can be read cleanly [5], [6]. Labels are sampled as $Y \sim \mathrm{Bernoulli}(1/2)$ and we write $S = 2Y - 1 \in \{-1, +1\}$. Conditional on S, features are generated as

$$x_1 \sim \mathcal{N}(1.05S, 0.90^2), \qquad x_2 \sim \mathcal{N}(1.25S + \delta(t), 0.55^2).$$

The first coordinate is a stable signal. The second is a nuisance coordinate that is predictive at training time but drifts during deployment. The label mechanism is held fixed throughout. Only $\delta(t)$ changes. The benchmark is intentionally simple. Its job is to make the paper's directional mechanism observable with as little confounding structure as possible.

The drift schedule is monotone and smooth, with speed

$$\begin{aligned} &\dot{\delta}(t) = 0.55 + 1.05e^{-((t-0.33)/0.10)^2} + 0.75e^{-((t-0.76)/0.08)^2}, \\ &\delta(0) = 0, \end{aligned}$$

so the stream contains two periods of rapid motion. We train a one-hidden-layer ReLU classifier with 12 hidden units, binary cross-entropy loss, Adam with learning rate 0.02, and 140 epochs. Each training set has 1024 examples drawn at $t = 0$, each deployment time uses a fresh sample of 2048 points, and we sweep $\lambda \in \{0, 0.01, 0.03, 0.08\}$ over two random seeds. DTR uses the true drift direction $V = e_2$ so that this experiment isolates the theorem-to-method link rather than subspace-estimation error.

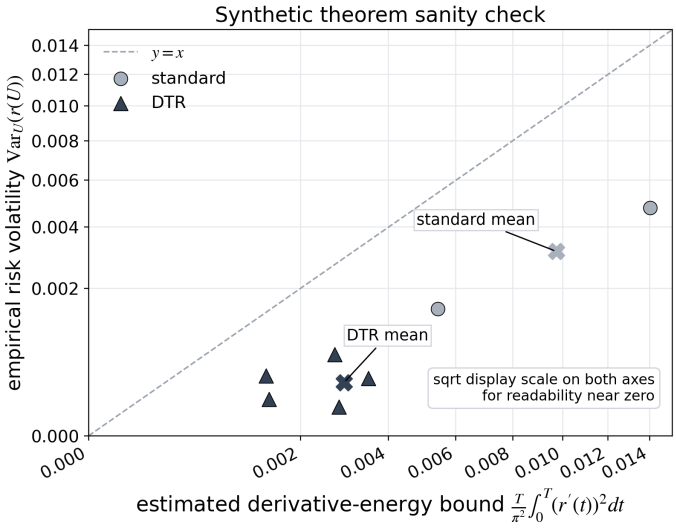

Fig. 2. Synthetic time-domain sanity check. Each point is one trained model evaluated over the drifting deployment path. The horizontal axis is the derivative-energy quantity from Lemma 1 and (3), and the vertical axis is empirical volatility $\mathrm{Var}_U(r(U))$. DTR shifts models toward lower energy and lower volatility.

Figure 2 plays the lemma-linked role in the paper's narrative arc. Each point is one trained model, plotted by empirical risk volatility in (2) versus the derivative-energy quantity from Lemma 1 and (3). The points sit below the diagonal, as they should, and DTR moves the cloud toward the lower-left corner. The saved summaries also respect the full Jacobian-velocity upper bound from Theorem 1, but that bound is numerically looser, so the figure emphasizes the sharper temporal step that anchors the rest of the pipeline. Across all eight models, mean volatility falls from 3.13×10^{-3} to 2.62×10^{-4}, while mean directional gain falls from 44.9 to 1.91.

B. Directional versus isotropic smoothing

The theorem suggests a sharper empirical question than whether "Jacobian regularization helps." When drift is concentrated in one known direction, does directional smoothing beat isotropic smoothing? This is the paper's main algorithmic comparison, because it asks whether the low-rank reduction is merely descriptive or whether it really changes the design choice. We answer that question in the same synthetic environment by replacing the DTR penalty with an isotropic Jacobian penalty

$$\lambda \mathbb{E}\|\nabla f_\theta(X)\|^2,$$

while keeping the directional penalty

$$\lambda \mathbb{E}\|J_f(X)V\|_F^2$$

for DTR, with the same architecture, optimizer, seeds, and sweep $\lambda \in \{0.01, 0.03, 0.08\}$. Under rank-1 drift, isotropic smoothing spends budget in directions the deployment path does not substantially traverse, whereas DTR spends that budget only along the realized motion. Table I summarizes the mean values across the nonzero-λ sweep. DTR improves on both baselines on every metric. Figure 3 (left) shows the matched $\lambda = 0.03$ comparison.

TABLE I
DIRECTIONAL COMPARISON SHOWING THE MEAN OVER THE NONZERO-λ SWEEP. LOWER IS BETTER ON ALL METRICS.

Method	Deriv. energy	Volatility	Dir. gain	Term. risk
Standard	4.80×10^{-3}	3.13×10^{-3}	44.9	0.171
Isotropic	6.11×10^{-4}	4.46×10^{-4}	2.15	0.168
DTR	3.14×10^{-4}	2.62×10^{-4}	1.91	0.134

We then test drift-subspace misspecification by rotating the one-dimensional subspace used by DTR. Writing the estimated direction as

$$\hat{v}_\alpha = (\sin\alpha, \cos\alpha),$$

its alignment with the true drift direction e_2 is $\cos\alpha$. With the correct subspace, the directional advantage remains strongest. With a mild 20° rotation, volatility rises by a factor of 1.43 and terminal risk by a factor of 1.20 relative to aligned DTR, but both remain far below the standard model. The same cosine-sine bookkeeping used in Proposition 1 helps explain that pattern: small angular error mainly rescales the aligned contribution, whereas large misalignment transfers weight to orthogonal energy. With an orthogonal subspace, the effect largely disappears. Derivative energy rises by a factor of 31.7, volatility by 23.9, and directional gain by 19.4. Figure 3 (right) therefore sharpens the paper's claim in the intended way. DTR is more targeted than isotropic Jacobian smoothing when the drift geometry is right, and its gains depend on keeping the estimated drift subspace reasonably aligned with realized motion.

C. Air Quality field deployment

We next test whether the same directional story survives on a real field dataset with documented drift. The role of this experiment is different from the synthetic studies. It is not a clean theorem check or a controlled comparison, but a test of whether the same geometry still leaves a measurable trace once the drift subspace must be estimated from data. The UCI Air Quality dataset records hourly outputs of a five-sensor gas array together with temperature and humidity in an Italian city, and the associated field-study paper explicitly notes both concept drift and sensor drift [23]. We predict the reference CO concentration $\mathrm{CO(GT)}$ from the five sensor channels $\mathrm{PT08.S1}$–$\mathrm{PT08.S5}$ together with T, RH, and AH. Rows with the dataset's -200 missing-value marker are removed, leaving 1573 training points from the first 12 weeks, 580 validation points from the next 4 weeks, and 5191 deployment points from the remaining horizon, grouped into 20 biweekly deployment blocks.

To keep the protocol aligned with the theorem, the predictor is frozen after training and deployment risk is the blockwise mean-squared error trajectory. We estimate a fixed two-dimensional drift subspace V from unlabeled deployment covariates by taking the top singular vectors of consecutive

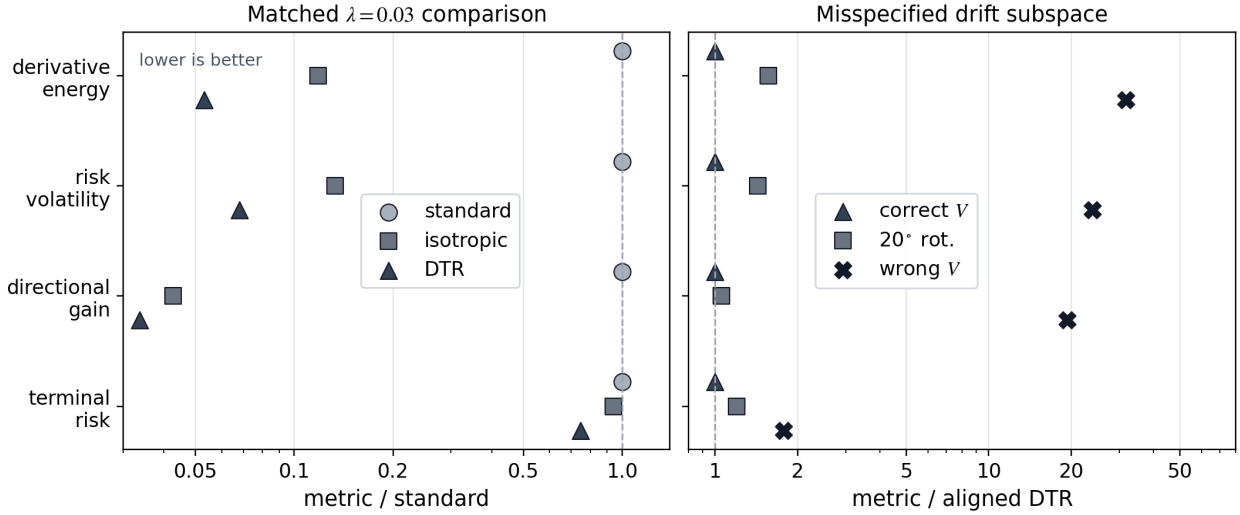

Fig. 3. Synthetic directional comparisons. Left shows matched $\lambda = 0.03$ ratios for derivative energy, risk volatility, directional gain, and terminal risk, normalized by the standard model. DTR is lower than isotropic Jacobian regularization on all four metrics. Right shows the same metrics normalized by aligned DTR for the correct subspace, a mild 20° rotation, and a wrong subspace. Mild misspecification is tolerable, but wrong subspaces are not.

biweekly mean-shift vectors. We then train a two-hidden-layer ReLU regressor with width 32, Adam with learning rate 0.01, and $\lambda \in \{0, 0.003, 0.01, 0.03, 0.08\}$ over three random seeds. Monitoring uses the same geometry as the method, specialized to the blockwise rank-1 setting covered by Proposition 1. If μ_t is the standardized covariate mean in block t and $\Delta = 1$ block, then

$$s_t = \|\mu_t - \mu_{t-1}\|, \qquad v_t = \frac{\mu_t - \mu_{t-1}}{\|\mu_t - \mu_{t-1}\|},$$
$$g_t = \mathbb{E}[(\nabla f_\theta(X_t)^\top v_t)^2], \qquad h_t = s_t^2 g_t.$$

Because the block width is fixed, writing s_t without an explicit division by Δ only rescales the hazard score by a constant and therefore does not change the ranking interpretation used in this section.

The real-data result supports the same mechanism, even though the theorem is not claimed as an exact model of this field dataset. Across the full nonzero-λ sweep, mean deployment-risk volatility falls from 9.29×10^{-2} to 8.16×10^{-2}, while mean directional gain falls from 8.65×10^{-2} to 6.80×10^{-2}. The representative moderate setting $\lambda = 0.03$ is stronger. Across three seeds, mean volatility falls to 6.49×10^{-2} and mean directional gain to 6.09×10^{-2}, with lower mean terminal risk (0.149 versus 0.169). Figure 4 shows one representative seed. The standard regressor begins at blockwise MSE 0.363, peaks at 1.15, and ends at 0.176. The DTR regressor begins at 0.342, peaks at 0.949, and ends lower at 0.152. Its mean hazard score also falls from 0.212 to 0.148, indicating that the same directional quantity used in training remains informative at monitoring time.

VII. Discussion and Limitations

The crucial caution is Assumption A3. Dynamic covariate shift by itself does not imply that the conditional-risk field is controlled by the score Jacobian along the realized path. We impose that domination explicitly because it is exactly what the theorem uses. The result is also not tied to ReLU networks in particular. ReLU models are one admissible class when deployment avoids kink sets almost surely for almost every deployment time, but smooth models fit the same theorem just as well.

The three empirical pieces play different roles. The time-domain sanity check asks whether the Poincare step is visible in trained models at all. The isotropic-versus-directional comparison asks whether the low-rank specialization has real design content. The Air Quality study asks whether the same directional quantities still matter once the setting becomes noisy and the drift subspace must be estimated. That progression is targeted evidence for the claim being made here rather than an attempt at broad benchmark coverage.

The low-rank story also depends on estimating a drift subspace. The synthetic misspecification study suggests that moderate angular error is tolerable but large misalignment is not. In the Air Quality study we estimate a fixed subspace from unlabeled future covariates, which is more realistic than the aligned synthetic setting but still retrospective. A real online deployment would require rolling estimation of V_t, and the quality of the monitoring score would then depend on residual drift energy and subspace-estimation error. Finally, the Jacobian-velocity bound is an upper bound. It is useful because it is actionable, not because it is tight in every regime, and it is best interpreted as a design guide for regularization and monitoring rather than as a complete description of all failure modes.

VIII. Conclusion

Long-horizon robustness under dynamic shift is best viewed as a tangent-control problem. The data stream contributes a velocity, the model contributes a local linear map, and deployment volatility is governed by their interaction. The central message is simple. Deployment becomes unstable when

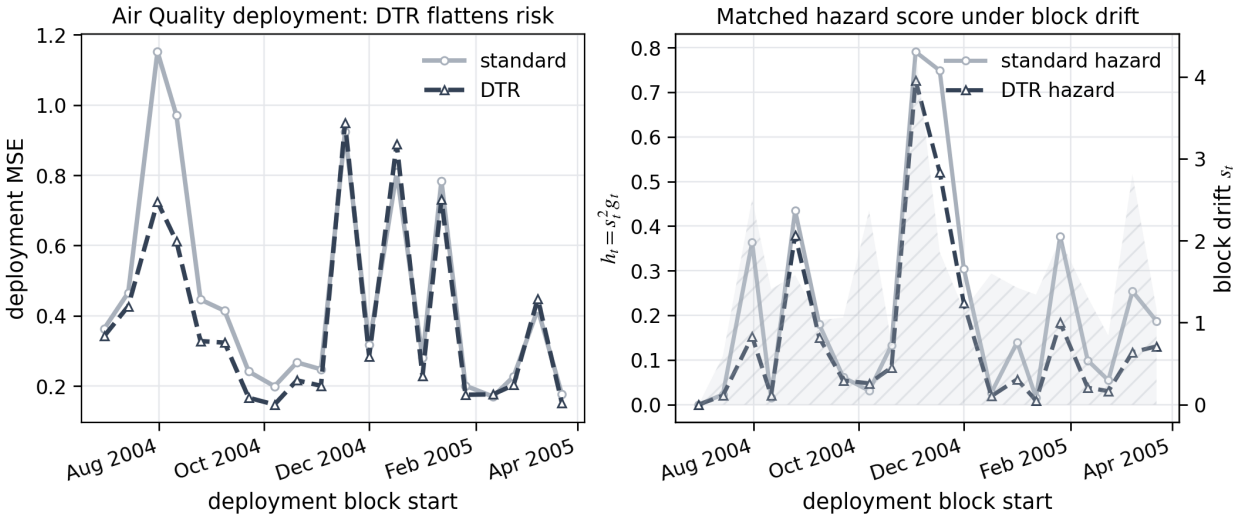

Fig. 4. Real field deployment on UCI Air Quality. Left shows biweekly deployment MSE for a standard CO regressor and a DTR regressor after training on the first 12 weeks and freezing the model. Right shows the matched hazard score $h_t = s_t^2 g_t$, with block drift magnitude in the background. DTR suppresses tangent gain along realized motion and therefore flattens both risk and hazard.

environmental motion repeatedly passes through directions to which the model is locally sensitive. The theorem package makes this precise, DTR acts on it, and the three experiments test whether the same directional geometry remains operative from clean synthetic settings through to noisy real sensor-drift data — it does, which is the paper's main empirical finding.

That positioning matters. Relative to temporal-shift benchmarks and static shift analyses [1], [4]–[6], the paper contributes a theorem-backed mechanism: deployment trajectories differ across frozen models because the same environmental motion can encounter very different directional tangent gains. Relative to test-time adaptation [10]–[12], it studies the frozen-model regime in which that geometry can be analyzed directly. Relative to Jacobian smoothing and shift detection [13], [15], [16], [22], it ties training-time regularization and deployment-time monitoring to the same directional notion of drift sensitivity. If future deployment motion is concentrated in a few directions, then robustness should be enforced, measured, and monitored in those directions rather than spread uniformly over the ambient space.

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